

# Boltzmann Calculation

## I. TRANSITION RATE

We start from the Hamiltonian

$$H_{\text{int}} = \frac{4}{U} \sum_{\substack{\langle i,j,k \rangle \\ \alpha\beta}} g_{ij} g_{kj} [1 - \lambda(X_{ij} + X_{kj})] p_{i\alpha}^\dagger (\vec{S}_j \cdot \vec{\sigma}_{\alpha\beta}) p_{k\beta} \quad (1)$$

First we fourier transform the Hamiltonian using

$$p_{i\alpha}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\mathbf{R}_i} p_{\mathbf{k}\alpha}^\dagger \quad (2)$$

$$\vec{S}_j = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}} e^{-i\mathbf{q}\mathbf{R}_j} \vec{S}_{\mathbf{q}} \quad (3)$$

For the first term in the interaction Hamiltonian, we have

$$H_{\text{int}} = H_{\text{int}}^I + H_{\text{int}}^{II}$$

where we have:

$$\begin{aligned} H_{\text{int}}^I &= \frac{1}{N^{3/2}} \frac{4}{U} \sum_{\mathbf{k}\mathbf{k}'\mathbf{q}} \sum_{\substack{\langle i,j,k \rangle \\ \alpha\beta}} g_{i0} g_{k0} e^{-i\mathbf{q}\mathbf{R}_j - i\mathbf{k}(\mathbf{R}_j + \mathbf{R}_i) + i\mathbf{k}'(\mathbf{R}_j + \mathbf{R}_k)} p_{\mathbf{k}\alpha}^\dagger \left[ \vec{S}_{\mathbf{q}} \cdot \vec{\sigma}_{\alpha\beta} \right] p_{\mathbf{k}'\beta} \\ &= \frac{1}{\sqrt{N}} \sum_{ik} \sum_{\substack{\mathbf{k}\mathbf{k}' \\ \alpha\beta}} g_{ik}^I(\mathbf{k}, \mathbf{k}') p_{\mathbf{k}\alpha}^\dagger \left[ \vec{S}_{\mathbf{k}-\mathbf{k}'} \cdot \vec{\sigma}_{\alpha\beta} \right] p_{\mathbf{k}'\beta} \end{aligned}$$

and we defined the form factor

$$g_{ik}(\mathbf{k}, \mathbf{k}') = \frac{4}{U} g_{i0} g_{k0} e^{-i\mathbf{k}\mathbf{R}_i + i\mathbf{k}'\mathbf{R}_k}$$

If we do the same for the phonon part

$$\begin{aligned} H_{\text{int}}^{II} &= -\lambda \frac{1}{N^2} \frac{4}{U} \sum_{\mathbf{k}\mathbf{k}'\mathbf{q}\mathbf{q}'} \sum_{\substack{\langle i,j,k \rangle \\ \alpha\beta}} g_{ij} g_{kj} (X_i(\mathbf{q}') + X_k(\mathbf{q}')) e^{-i\mathbf{q}\mathbf{R}_j - i\mathbf{k}\mathbf{R}_i + i\mathbf{k}'\mathbf{R}_k - i\mathbf{q}'\mathbf{R}_j} p_{\mathbf{k}\alpha}^\dagger \left[ \vec{S}_{\mathbf{q}} \cdot \vec{\sigma}_{\alpha\beta} \right] p_{\mathbf{k}'\beta} \\ &= -\frac{\lambda}{N} \sum_{\mathbf{k}\mathbf{k}'\mathbf{q}} \sum_{\substack{ik \\ \alpha\beta}} g_{ik}(\mathbf{k}, \mathbf{k}') (X_i(\mathbf{k} - \mathbf{k}' - \mathbf{q}) + X_k(\mathbf{k} - \mathbf{k}' - \mathbf{q})) p_{\mathbf{k}\alpha}^\dagger \left[ \vec{S}_{\mathbf{q}} \cdot \vec{\sigma}_{\alpha\beta} \right] p_{\mathbf{k}'\beta} \end{aligned}$$

We now define

$$V_{\mathbf{k}\mathbf{k}'}^{\alpha\beta} = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}} \sum_{ik} g_{ik}(\mathbf{k}, \mathbf{k}') \left( \delta(\mathbf{q} - \mathbf{k} + \mathbf{k}') - \frac{\lambda}{\sqrt{N}} (X_i(\mathbf{k} - \mathbf{k}' - \mathbf{q}) + X_k(\mathbf{k} - \mathbf{k}' - \mathbf{q})) \right) \left[ \vec{S}_{\mathbf{q}} \cdot \vec{\sigma}_{\alpha\beta} \right] \quad (4)$$

$$= U_{\mathbf{k}\mathbf{k}'}^{\alpha\beta} + W_{\mathbf{k}\mathbf{k}'}^{\alpha\beta} \quad (5)$$

Then we have

$$H_{\text{int}} = \sum_{\mathbf{k}\mathbf{k}'\alpha\beta} V_{\mathbf{k}\mathbf{k}'}^{\alpha\beta} p_{\mathbf{k}\alpha}^\dagger p_{\mathbf{k}'\beta} \quad (6)$$

Then, we can use usual time dependent perturbation theory to estimate the decay rate from a state of momentum  $\mathbf{k}$  and spin  $\alpha$  to a state of momentum  $\mathbf{k}'$  and spin  $\beta$  as follows:

$$\mathcal{P}_{\mathbf{k}\alpha \rightarrow \mathbf{k}'\beta}(t) \sim \int_{-\infty}^t dt' dt'' \overline{V_{\mathbf{k}\mathbf{k}'}^{\alpha\beta}(t')} V_{\mathbf{k}\mathbf{k}'}^{\alpha\beta}(t'') e^{i(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(t' - t'')}.$$

However, we want the total transition probability when  $t \rightarrow \infty$ . Additionally, we take the thermal expextation value over the phonon and the spin subsystems. This reduces the number of terms to be evaluated since  $\langle X \rangle = 0$ . As such we only have to consider

$$\mathcal{P}_{\mathbf{k}\alpha \rightarrow \mathbf{k}'\beta} = \int_{-\infty}^{\infty} dt' dt'' \left\langle \left\langle \overline{U_{\mathbf{k}\mathbf{k}'}^{\alpha\beta}(t')} U_{\mathbf{k}\mathbf{k}'}^{\alpha\beta}(t'') \right\rangle \right\rangle e^{i(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(t' - t'')} \quad (7)$$

$$+ \int_{-\infty}^{\infty} dt' dt'' \left\langle \left\langle \overline{W_{\mathbf{k}\mathbf{k}'}^{\alpha\beta}(t')} W_{\mathbf{k}\mathbf{k}'}^{\alpha\beta}(t'') \right\rangle \right\rangle e^{i(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(t' - t'')} \quad (8)$$

$$= \mathcal{P}_{\mathbf{k}\alpha \rightarrow \mathbf{k}'\beta}^I + \mathcal{P}_{\mathbf{k}\alpha \rightarrow \mathbf{k}'\beta}^{II} \quad (9)$$

We now evaluate the first line which does not involve phonon operators

$$\begin{aligned} \mathcal{P}_{\mathbf{k}\alpha \rightarrow \mathbf{k}'\beta}^I &= \frac{1}{N} \sum_{ik i' k'} \int_{-\infty}^{\infty} dt' dt'' g_{ik}(\mathbf{k}, \mathbf{k}') \overline{g_{i' k'}}(\mathbf{k}, \mathbf{k}') \left\langle \left\langle \left[ S_{\mathbf{k}-\mathbf{k}'}^{\mu}(t') \sigma_{\alpha\beta}^{\mu} \right] \left[ S_{\mathbf{k}-\mathbf{k}'}^{\nu}(t'') \sigma_{\alpha\beta}^{\nu} \right] \right\rangle \right\rangle e^{i(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(t' - t'')} \\ &= \frac{1}{N} \sum_{ik i' k'} \int_{-\infty}^{\infty} dt' dt'' g_{ik}(\mathbf{k}, \mathbf{k}') \overline{g_{i' k'}}(\mathbf{k}, \mathbf{k}') \left\langle \left\langle S_{\mathbf{k}-\mathbf{k}'}^{\mu}(t') \overline{S_{\mathbf{k}-\mathbf{k}'}^{\nu}(t'')} \right\rangle \right\rangle \sigma_{\alpha\beta}^{\mu} \sigma_{\beta\alpha}^{\nu} e^{i(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(t' - t'')} \end{aligned}$$

Taking the ensemble average yields the dynamic structure factor:

$$\begin{aligned} \mathcal{P}_{\mathbf{k}\alpha \rightarrow \mathbf{k}'\beta}^I &= \frac{1}{N} \sum_{ik i' k'} \int_{-\infty}^{\infty} dt' dt'' g_{ik}(\mathbf{k}, \mathbf{k}') \overline{g_{i' k'}}(\mathbf{k}, \mathbf{k}') S^{\mu\nu}(\mathbf{k} - \mathbf{k}', t' - t'') \sigma_{\alpha\beta}^{\mu} \sigma_{\beta\alpha}^{\nu} e^{i(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(t' - t'')} \\ &= \frac{1}{N} \sum_{ik i' k'} g_{ik}^I(\mathbf{k}, \mathbf{k}') \overline{g_{i' k'}}(\mathbf{k}, \mathbf{k}') S^{\mu\nu}(\mathbf{k} - \mathbf{k}', \varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'}) \sigma_{\alpha\beta}^{\mu} \sigma_{\beta\alpha}^{\nu} \end{aligned}$$

In taking the fourier transform we used sleight of hand since formally we make the change of variables  $t = t' - t''$  and  $\tau = \frac{1}{2}(t' + t'')$ . The integral over  $\tau$  is irrelevant because we are interested in the transition rate,. This integral gives an overall factor of  $\tau$  which is then canceled by a derivative. We therefore use interchangeably transition probability and transition rate in what follows. Tracing over the spin we get

$$\text{tr} \sigma^{\mu} \sigma^{\nu} = \delta^{\mu\nu} \quad (10)$$

Which then yields the transition rate given by

$$\begin{aligned} \mathcal{P}_{\mathbf{k} \rightarrow \mathbf{k}'}^I &= \frac{1}{N} \sum_{ik i' k'} g_{ik}(\mathbf{k}, \mathbf{k}') \overline{g_{i' k'}}(\mathbf{k}, \mathbf{k}') S(\mathbf{k} - \mathbf{k}', \varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'}) \\ &= \Lambda_I(\mathbf{k}, \mathbf{k}') S(\mathbf{k} - \mathbf{k}', \varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'}) \end{aligned}$$

where we defined the form factor

$$\Lambda_I(\mathbf{k}, \mathbf{k}') = \frac{1}{N} \sum_{ik i' k'} g_{ik}(\mathbf{k}, \mathbf{k}') \overline{g_{i' k'}}(\mathbf{k}, \mathbf{k}') \quad (11)$$

For the second term we have

$$\begin{aligned}
\mathcal{P}_{\mathbf{k}\alpha \rightarrow \mathbf{k}'\beta}^{II} &= \frac{\lambda^2}{N^2} \sum_{i\mathbf{k}i'\mathbf{k}'} \sum_{\mathbf{q}\mathbf{q}'} \int_{-\infty}^{\infty} dt' dt'' g_{ik}(\mathbf{k}, \mathbf{k}') \overline{g_{i'k'}}(\mathbf{k}, \mathbf{k}') \\
&\quad \left\langle \left\langle \left[ (X_i(\mathbf{k} - \mathbf{k}' - \mathbf{q}, t') + X_k(\mathbf{k} - \mathbf{k}' - \mathbf{q}, t')) S_{\mathbf{q}}^{\mu}(t') \right] \overline{\left[ (X_{i'}(\mathbf{k} - \mathbf{k}' - \mathbf{q}', t'') + X_{k'}(\mathbf{k} - \mathbf{k}' - \mathbf{q}', t'')) S_{\mathbf{q}'}^{\nu}(t'') \right]} \right\rangle \right\rangle \\
&\quad \sigma_{\alpha\beta}^{\mu} \sigma_{\beta\alpha}^{\nu} e^{i(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(t' - t'')} \\
&= \frac{\lambda^2}{N^2} \sum_{i\mathbf{k}i'\mathbf{k}'} \sum_{\mathbf{q}\mathbf{q}'} \int_{-\infty}^{\infty} dt' dt'' g_{ik}(\mathbf{k}, \mathbf{k}') \overline{g_{i'k'}}(\mathbf{k}, \mathbf{k}') \\
&\quad \left\langle \left\langle (X_i(\mathbf{k} - \mathbf{k}' - \mathbf{q}, t') + X_k(\mathbf{k} - \mathbf{k}' - \mathbf{q}, t')) \overline{(X_{i'}(\mathbf{k} - \mathbf{k}' - \mathbf{q}', t'') + X_{k'}(\mathbf{k} - \mathbf{k}' - \mathbf{q}', t''))} \right\rangle \right\rangle \\
&\quad \left\langle \left\langle S_{\mathbf{q}}^{\mu}(t') \overline{S_{\mathbf{q}'}^{\nu}(t'')} \right\rangle \right\rangle \sigma_{\alpha\beta}^{\mu} \sigma_{\beta\alpha}^{\nu} e^{i(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(t' - t'')}
\end{aligned}$$

Using the leading order correlator for the phonons yields a delta function in momentum that cancels one of the sums above:

$$\begin{aligned}
\mathcal{P}_{\mathbf{k}\alpha \rightarrow \mathbf{k}'\beta}^{II} &= \frac{\lambda^2}{N^2} \sum_{i\mathbf{k}i'\mathbf{k}'} \sum_{\mathbf{q}} \int_{-\infty}^{\infty} dt' dt'' g_{ik}(\mathbf{k}, \mathbf{k}') \overline{g_{i'k'}}(\mathbf{k}, \mathbf{k}') \\
&\quad \left\langle \left\langle (X_i(\mathbf{k} - \mathbf{k}' - \mathbf{q}, t') + X_k(\mathbf{k} - \mathbf{k}' - \mathbf{q}, t')) \overline{(X_{i'}(\mathbf{k} - \mathbf{k}' - \mathbf{q}, t'') + X_{k'}(\mathbf{k} - \mathbf{k}' - \mathbf{q}, t''))} \right\rangle \right\rangle \\
&\quad S^{\mu\nu}(\mathbf{k} - \mathbf{k}', t' - t'') \sigma_{\alpha\beta}^{\mu} \sigma_{\beta\alpha}^{\nu} e^{i(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(t' - t'')}
\end{aligned}$$

Taking the trace over spin

$$\begin{aligned}
\mathcal{P}_{\mathbf{k} \rightarrow \mathbf{k}'}^{II} &= \frac{\lambda^2}{N^2} \sum_{i\mathbf{k}i'\mathbf{k}'} \sum_{\mathbf{q}} \int_{-\infty}^{\infty} dt' dt'' g_{ik}(\mathbf{k}, \mathbf{k}') \overline{g_{i'k'}}(\mathbf{k}, \mathbf{k}') \\
&\quad \left\langle \left\langle (X_i(\mathbf{k} - \mathbf{k}' - \mathbf{q}, t') + X_k(\mathbf{k} - \mathbf{k}' - \mathbf{q}, t')) \overline{(X_{i'}(\mathbf{k} - \mathbf{k}' - \mathbf{q}, t'') + X_{k'}(\mathbf{k} - \mathbf{k}' - \mathbf{q}, t''))} \right\rangle \right\rangle \\
&\quad S(\mathbf{k} - \mathbf{k}', t' - t'') e^{i(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(t' - t'')}
\end{aligned}$$

we can then use the finite temperature correlator

$$\left\langle \left\langle X_i(\mathbf{q}, t) \overline{X_j(\mathbf{q}, t')} \right\rangle \right\rangle = \frac{\hbar \delta_{ij}}{2M\omega_{\mathbf{q}}} [1 + n_B(\hbar\omega_{\mathbf{q}})] e^{i\omega_{\mathbf{q}}(t-t')} \quad (12)$$

where we assumed that the oscillations along the different bond directions have the same dispersion.

$$\begin{aligned}
\mathcal{P}_{\mathbf{k} \rightarrow \mathbf{k}'}^{II} &= \frac{\lambda^2}{N^2} \sum_{i\mathbf{k}i'\mathbf{k}'} \sum_{\mathbf{q}} \int_{-\infty}^{\infty} dt' dt'' g_{ik}(\mathbf{k}, \mathbf{k}') \overline{g_{i'k'}}(\mathbf{k}, \mathbf{k}') (\delta_{ii'} + \delta_{ik'} + \delta_{ki'} + \delta_{kk'}) \\
&\quad \frac{\hbar}{2M\omega_{\mathbf{q}}} [1 + n_B(\hbar\omega_{\mathbf{q}})] e^{i\omega_{\mathbf{q}}(t-t')} S(\mathbf{k} - \mathbf{k}', t' - t'') e^{i(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(t' - t'')} \\
&= \lambda^2 \sum_{\mathbf{q}} \int_{-\infty}^{\infty} dt' dt'' \Lambda_{II}(\mathbf{k}, \mathbf{k}') \frac{\hbar}{2M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} [1 + n_B(\hbar\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}})] e^{i\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}(t-t')} S(\mathbf{k} - \mathbf{k}', t' - t'') e^{i(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(t' - t'')} \\
&= \lambda^2 \sum_{\mathbf{q}} \int_{-\infty}^{\infty} dt' dt'' \Lambda_{II}(\mathbf{k}, \mathbf{k}') \frac{\hbar}{2M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} [1 + n_B(\hbar\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}})] e^{i\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}(t-t')} S(\mathbf{k} - \mathbf{k}', t' - t'') e^{i(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(t' - t'')} \\
&= \lambda^2 \sum_{\mathbf{q}} \Lambda_{II}(\mathbf{k}, \mathbf{k}') \frac{\hbar}{2M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} [1 + n_B(\hbar\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}})] S(\mathbf{k} - \mathbf{k}', \varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}})
\end{aligned}$$

where we defined the form factor

$$\Lambda_{II}(\mathbf{k}, \mathbf{k}') = \frac{1}{N} \sum_{i k i' k'} g_{ik}(\mathbf{k}, \mathbf{k}') \overline{g_{i'k'}}(\mathbf{k}, \mathbf{k}') (\delta_{ii'} + \delta_{ik'} + \delta_{ki'} + \delta_{kk'}) \quad (13)$$

As such, we have

$$\mathcal{P}_{\mathbf{k} \rightarrow \mathbf{k}'} = \mathcal{P}_{\mathbf{k} \rightarrow \mathbf{k}'}^I + \mathcal{P}_{\mathbf{k} \rightarrow \mathbf{k}'}^{II} \quad (14)$$

$$= \Lambda_I(\mathbf{k}, \mathbf{k}') S(\mathbf{k} - \mathbf{k}', \varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'}) \quad (15)$$

$$+ \sum_{\mathbf{q}} \Lambda_{II}(\mathbf{k}, \mathbf{k}') \frac{\hbar \lambda^2}{2M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} [1 + n_B(\hbar\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}})] S(\mathbf{k} - \mathbf{k}', \varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) \quad (16)$$

In the high temperature limit where the phonons are in the equipartition regime, we have a transition rate given by

$$\mathcal{P}_{\mathbf{k} \rightarrow \mathbf{k}'} = \Lambda_I(\mathbf{k}, \mathbf{k}') S(\mathbf{k} - \mathbf{k}', \varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'}) + \frac{k_b T}{2} \lambda^2 \sum_{\mathbf{q}} \Lambda_{II}(\mathbf{k}, \mathbf{k}') \frac{S(\mathbf{k} - \mathbf{k}', \varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}})}{M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}^2}. \quad (17)$$

## II. COLLISION INTEGRAL

This transition rate enters the collision integral in the Boltzmann formalism as follows

$$\begin{aligned} \left( \frac{\partial f}{\partial t} \right)_{\text{col}} &= \sum_{\mathbf{k}'} \mathcal{P}_{\mathbf{k} \rightarrow \mathbf{k}'} f(\mathbf{k}) (1 - f(\mathbf{k}')) - \mathcal{P}_{\mathbf{k}' \rightarrow \mathbf{k}} f(\mathbf{k}') (1 - f(\mathbf{k})) \\ &= \sum_{\mathbf{k}'} \mathcal{P}_{\mathbf{k}\mathbf{k}'} [f(\mathbf{k}) (1 - f(\mathbf{k}')) - f(\mathbf{k}') (1 - f(\mathbf{k}))] \\ &= \sum_{\mathbf{k}'} \mathcal{P}_{\mathbf{k}\mathbf{k}'} [f(\mathbf{k}) - f(\mathbf{k}')] \end{aligned}$$

Having a concrete form for the collision integral we display the full Boltzmann equation

$$\frac{\partial f}{\partial t} + \frac{\partial \mathbf{k}}{\partial t} \cdot \nabla_{\mathbf{k}} f + \mathbf{v} \cdot \nabla_{\mathbf{r}} f = \left( \frac{\partial f}{\partial t} \right)_{\text{col}}. \quad (18)$$

we will consider the situation where there is no explicit time or space dependence on the distribution function. In the absence of magnetic fields we then have

$$\frac{\partial \mathbf{k}}{\partial t} = -e\mathbf{E}.$$

With this, the Boltzmann equation reduces to

$$-e\mathbf{E} \cdot \nabla_{\mathbf{k}} f = \left( \frac{\partial f}{\partial t} \right)_{\text{col}}$$

To proceed, we linearize the Boltzmann equation

$$\begin{aligned} f(\mathbf{k}) &\approx n_F(\varepsilon_{\mathbf{k}}) - \frac{dn_F(\varepsilon_{\mathbf{k}})}{d\varepsilon_{\mathbf{k}}} \Phi(\mathbf{k}) \\ &= n_F(\varepsilon_{\mathbf{k}}) + n_F(\varepsilon_{\mathbf{k}}) (1 - n_F(\varepsilon_{\mathbf{k}})) \Phi(\mathbf{k}) \end{aligned}$$

Inserting this in the collision integral, we get

$$\begin{aligned} \sum_{\mathbf{k}'} W_{\mathbf{k}\mathbf{k}'} [f(\mathbf{k}) - f(\mathbf{k}')] &= \sum_{\mathbf{k}'} \mathcal{P}_{\mathbf{k}\mathbf{k}'} [n_F(\varepsilon_{\mathbf{k}}) + n_F(\varepsilon_{\mathbf{k}}) (1 - n_F(\varepsilon_{\mathbf{k}})) \Phi(\mathbf{k}) - n_F(\varepsilon_{\mathbf{k}'}) - n_F(\varepsilon_{\mathbf{k}'} (1 - n_F(\varepsilon_{\mathbf{k}')) \Phi(\mathbf{k}')] \\ &= n_F(\varepsilon_{\mathbf{k}}) (1 - n_F(\varepsilon_{\mathbf{k}})) \sum_{\mathbf{k}'} \mathcal{P}_{\mathbf{k}\mathbf{k}'} [\Phi(\mathbf{k}) - \Phi(\mathbf{k}')] \\ &= \left( -\frac{dn_F(\varepsilon_{\mathbf{k}})}{d\varepsilon_{\mathbf{k}}} \right) \sum_{\mathbf{k}'} \mathcal{P}_{\mathbf{k}\mathbf{k}'} [\Phi(\mathbf{k}) - \Phi(\mathbf{k}')] \end{aligned}$$

Where the last equality follows from the fact that  $W_{\mathbf{k}\mathbf{k}'} \propto \delta(\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})$ . Working to a consistent order in the electric field, we then have

$$\begin{aligned} -e\mathbf{E} \cdot \nabla_{\mathbf{k}} f &= -e\mathbf{E} \cdot \nabla_{\mathbf{k}} n_F \\ &= -e\mathbf{E} \cdot \nabla_{\mathbf{k}} \varepsilon_{\mathbf{k}} \frac{dn_F(\varepsilon_{\mathbf{k}})}{d\varepsilon_{\mathbf{k}}} \\ &= e\mathbf{E} \cdot \mathbf{v}_{\mathbf{k}} \left( -\frac{dn_F(\varepsilon_{\mathbf{k}})}{d\varepsilon_{\mathbf{k}}} \right) \end{aligned}$$

Consequently, the equation reduces to the form

$$\begin{aligned} e\mathbf{E} \cdot \mathbf{v}_{\mathbf{k}} &= \sum_{\mathbf{k}'} \mathcal{P}_{\mathbf{k}\mathbf{k}'} [\Phi(\mathbf{k}) - \Phi(\mathbf{k}')] \\ &= \sum_{\mathbf{k}'} (\mathcal{P}_{\mathbf{k} \rightarrow \mathbf{k}'}^I + \mathcal{P}_{\mathbf{k}' \rightarrow \mathbf{k}}^{II}) [\Phi(\mathbf{k}) - \Phi(\mathbf{k}')] . \end{aligned}$$

We could try to solve this equation numerically, however, we take a different approach here and use the relaxation-time approximation. In this approximation, we take

$$\left( \frac{\partial f}{\partial t} \right)_{\text{col}} = -\frac{f(\mathbf{k}) - n_F(\varepsilon_{\mathbf{k}})}{\tau(\mathbf{k})} .$$

As such, we have

$$\frac{f(\mathbf{k}) - n_F(\varepsilon_{\mathbf{k}})}{\tau(\mathbf{k})} = \left( -\frac{dn_F(\varepsilon_{\mathbf{k}})}{d\varepsilon_{\mathbf{k}}} \right) \sum_{\mathbf{k}'} \mathcal{P}_{\mathbf{k}\mathbf{k}'} [\Phi(\mathbf{k}) - \Phi(\mathbf{k}')] .$$

Now we use the following parametrization for

$$\left( -\frac{dn_F(\varepsilon_{\mathbf{k}})}{d\varepsilon_{\mathbf{k}}} \right) \Phi(\mathbf{k}) = \mathbf{k} \cdot \mathbf{E} \varphi(\varepsilon_{\mathbf{k}}) .$$

With this, we have

$$\frac{f(\mathbf{k}) - n_F(\varepsilon_{\mathbf{k}})}{\tau(\mathbf{k})} = \varphi(\varepsilon_{\mathbf{k}}) \sum_{\mathbf{k}'} \mathcal{P}_{\mathbf{k}\mathbf{k}'} [\mathbf{k} - \mathbf{k}'] \cdot \mathbf{E}$$

We use the following relations

$$\hat{\mathbf{k}} \cdot \hat{\mathbf{E}} = \cos \theta ,$$

$$\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' = \cos \theta' ,$$

we have then

$$\hat{\mathbf{k}}' \cdot \hat{\mathbf{E}} = \cos \theta \cos \theta' + \sin \theta \sin \theta' .$$

inserting this in the collision integral, we get

$$\begin{aligned} \sum_{\mathbf{k}'} \mathcal{P}_{\mathbf{k}\mathbf{k}'} [\mathbf{k} - \mathbf{k}'] \cdot \mathbf{E} &= |\mathbf{k}| |\mathbf{E}| V \int \frac{d\mathbf{k}'}{(2\pi)^2} \mathcal{P}_{\mathbf{k}\mathbf{k}'} (\cos \theta - \cos \theta \cos \theta' - \sin \theta \sin \theta') \\ &= \varphi(\varepsilon_{\mathbf{k}}) |\mathbf{k}| |\mathbf{E}| \cos \theta V \int \frac{d\mathbf{k}'}{(2\pi)^2} \mathcal{P}_{\mathbf{k}\mathbf{k}'} \left( 1 - \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' - \frac{\sin \theta}{\cos \theta} \hat{\mathbf{k}} \times \hat{\mathbf{k}}' \right) \\ &= \mathbf{k} \cdot \mathbf{E} \varphi(\varepsilon_{\mathbf{k}}) V \int \frac{d\mathbf{k}'}{(2\pi)^2} \mathcal{P}_{\mathbf{k}\mathbf{k}'} \left( 1 - \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' - \tan \theta \hat{\mathbf{k}} \times \hat{\mathbf{k}}' \right) \\ &= (f(\mathbf{k}) - n_F(\varepsilon_{\mathbf{k}})) V \int \frac{d\mathbf{k}'}{(2\pi)^2} \mathcal{P}_{\mathbf{k}\mathbf{k}'} \left( 1 - \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}' - \tan \theta \hat{\mathbf{k}} \times \hat{\mathbf{k}}' \right) \end{aligned}$$

we can argue that  $\mathbf{E} \cdot \mathbf{k} = Ek$  such that  $\theta = 0$  and

$$\sum_{\mathbf{k}'} \mathcal{P}_{\mathbf{k}\mathbf{k}'} [\mathbf{k} - \mathbf{k}'] \cdot \mathbf{E} = (f(\mathbf{k}) - n_F(\varepsilon_{\mathbf{k}})) V \int \frac{d\mathbf{k}'}{(2\pi)^2} \mathcal{P}_{\mathbf{k}\mathbf{k}'} (1 - \cos \theta')$$

where we implicitly assumed an isotropic dispersion. After all, we get the following expression for the relaxation time

$$\frac{1}{\tau(\mathbf{k})} = V \int \frac{d\mathbf{k}'}{(2\pi)^D} \mathcal{P}_{\mathbf{k} \rightarrow \mathbf{k}'} (1 - \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \quad (19)$$

$$= V \int \frac{d\mathbf{k}'}{(2\pi)^D} [\Lambda_I(\mathbf{k}, \mathbf{k}') S(\mathbf{k} - \mathbf{k}', \varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})] \quad (20)$$

$$+ \sum_{\mathbf{q}} \Lambda_{II}(\mathbf{k}, \mathbf{k}') \frac{\hbar \lambda^2}{2M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} [1 + n_B(\hbar\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}})] S(\mathbf{k} - \mathbf{k}', \varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}] (1 - \hat{\mathbf{k}} \cdot \hat{\mathbf{k}}') \quad (21)$$

We now calculate the in plane conductivity using this form for the relaxation time, we have to leading order

$$e\mathbf{E} \cdot \mathbf{v}_{\mathbf{k}} \left( -\frac{dn_F(\varepsilon_{\mathbf{k}})}{d\varepsilon_{\mathbf{k}}} \right) = \frac{f(\mathbf{k}) - n_F(\varepsilon_{\mathbf{k}})}{\tau(\mathbf{k})}.$$

Solving for the distribution function, we get

$$f(\mathbf{k}) = n_F(\varepsilon_{\mathbf{k}}) + e\tau(\mathbf{k}) \mathbf{E} \cdot \mathbf{v}_{\mathbf{k}} \left( -\frac{dn_F(\varepsilon_{\mathbf{k}})}{d\varepsilon_{\mathbf{k}}} \right).$$

The current can be calculated using the distribution function to leading order in  $\lambda$  and the Kondo coupling as

$$\mathbf{J} = en \langle \mathbf{v} \rangle = e \int \frac{d\mathbf{k}}{(2\pi)^D} v_{\mathbf{k}} f(\mathbf{k})$$

Since we have

$$e \int \frac{d\mathbf{k}}{(2\pi)^D} v_{\mathbf{k}} n_F(\varepsilon_{\mathbf{k}}) = 0,$$

the only contribution comes from the term proportional to the relaxation time

$$\mathbf{J} = e \int \frac{d\mathbf{k}}{(2\pi)^D} \tau(\mathbf{k}) (e\mathbf{E} \cdot \mathbf{v}_{\mathbf{k}}) \mathbf{v}_{\mathbf{k}} \left( -\frac{dn_F(\varepsilon_{\mathbf{k}})}{d\varepsilon_{\mathbf{k}}} \right).$$

Assuming the electric field is applied in the  $x$  direction, and we are interested in the  $x$  component of the current, we get

$$J_x = \left[ e^2 \int \frac{d\mathbf{k}}{(2\pi)^D} \tau(\mathbf{k}) v_{x,\mathbf{k}}^2 \left( -\frac{dn_F(\varepsilon_{\mathbf{k}})}{d\varepsilon_{\mathbf{k}}} \right) \right] E.$$

Then, the conductivity will be given by

$$\sigma_{\parallel} = e^2 \int \frac{d\mathbf{k}}{(2\pi)^D} \tau(\mathbf{k}) v_{x,\mathbf{k}}^2 \left( -\frac{dn_F(\varepsilon_{\mathbf{k}})}{d\varepsilon_{\mathbf{k}}} \right). \quad (22)$$

### III. CURRENT OPERATOR

Starting from the Hamiltonian

$$H = -t_p \sum_{ij\sigma} p_{i\sigma}^\dagger p_{j\sigma} + \frac{4t_c^2}{U} \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + \frac{4}{U} \sum_{ijk\alpha} \tilde{g}_{ij} \tilde{g}_{kj} p_{i\sigma}^\dagger (\mathbf{S}_j \cdot \boldsymbol{\sigma}_{\sigma\beta}) p_{k\beta} \quad (23)$$

$$= - \sum_{ij\sigma} t_{ij}^p p_{i\sigma}^\dagger p_{j\sigma} + \frac{4t_c^2}{U} \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + \sum_{ij\alpha} \tilde{g}_{ij} p_{i\alpha}^\dagger \left( \frac{2}{U} \sum_{k\beta} \tilde{g}_{kj} (\mathbf{S}_j \cdot \boldsymbol{\sigma}_{\alpha\beta}) p_{k\beta} \right) + \sum_{kj\beta} \tilde{g}_{kj} \left[ \sum_{i\alpha} \frac{2}{U} \tilde{g}_{ij} p_{i\alpha}^\dagger (\mathbf{S}_j \cdot \boldsymbol{\sigma}_{\alpha\beta}) \right] p_{k\beta} \quad (24)$$

$$= - \sum_{ij\sigma} t_{ij}^p p_{i\sigma}^\dagger p_{j\sigma} + \frac{4t_c^2}{U} \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + \sum_{\langle ij \rangle \alpha} \tilde{g}_{ij} p_{i\alpha}^\dagger c_{j\alpha} + \sum_{\langle ij \rangle \alpha} \tilde{g}_{ij} c_{j\alpha}^\dagger p_{i\alpha} \quad (25)$$

where we defined

$$\tilde{g}_{kj} = g_{kj} (1 - \lambda X_{kj})$$

and

$$c_{j\alpha}^\dagger = \sum_{i\alpha} \frac{2}{U} \tilde{g}_{ij} p_{i\alpha}^\dagger (\mathbf{S}_j \cdot \boldsymbol{\sigma}_{\alpha\beta}).$$

We want to get the current operator from the continuity equation.

$$i\partial_t n_{i\sigma}^p = [n_{i\sigma}^p, H] \quad (26)$$

Then we interpret the particle current as

$$J_N^\sigma = -i \sum_i \mathbf{R}_i [n_{i\sigma}^p, H] \quad (27)$$

$$= -i \sum_{ir\alpha} \mathbf{R}_i [n_{i\sigma}^p, h_{r\alpha}] \quad (28)$$

With the local hamiltonian that has a finite contribution to the current is given by:

$$h_{r\alpha} = \sum_l \frac{t_{rl}}{2} p_{r\alpha}^\dagger p_{l\alpha} + \frac{t_{rl}}{2} p_{l\alpha}^\dagger p_{r\alpha} + \tilde{g}_{rl} p_{r\alpha}^\dagger c_{l\alpha} + \tilde{g}_{rl} c_{l\alpha}^\dagger p_{r\alpha} \quad (29)$$

We do this for the current per spin, since it is not a priori obvious to me that the hamiltonian preserves spin. Since the density commutes with the Heisenberg term, we only need to calculate the commutator with the kinetic term

$$\left[ n_{i\sigma}^p, \frac{t_{rl}}{2} p_{r\alpha}^\dagger p_{l\alpha} \right] = \frac{t_{rl}}{2} [n_{i\sigma}, p_{r\alpha}^\dagger p_{l\alpha}] \quad (30)$$

$$= \frac{t_{rl}}{2} \delta_{\sigma\alpha} (\delta_{ri} - \delta_{li}) p_{r\alpha}^\dagger p_{l\alpha} \quad (31)$$

$$\left[ n_{i\sigma}^p, \frac{t_{rl}}{2} p_{l\alpha}^\dagger p_{r\alpha} \right] = \frac{t_{rl}}{2} \delta_{\sigma\alpha} (\delta_{li} - \delta_{ri}) p_{l\alpha}^\dagger p_{r\alpha} \quad (32)$$

Where we used

$$[n_{i\sigma}^p, p_{l\alpha}^\dagger] = \delta_{\sigma\alpha} \delta_{li} p_{l\alpha}^\dagger \quad (33)$$

and similarly

$$[n_{i\sigma}^p, p_{l\alpha}] = -\delta_{\sigma\alpha}\delta_{li}p_{l\alpha} \quad (34)$$

Then when we calculate the relevant commutators, we get

$$[n_{i\sigma}^p, \tilde{g}_{rl}p_{r\alpha}^\dagger c_{l\alpha}] = \frac{2}{U}\tilde{g}_{rl} \sum_{k\beta} \tilde{g}_{kl} (\mathbf{S}_l \cdot \boldsymbol{\sigma}_{\alpha\beta}) [n_{i\sigma}^p, p_{r\alpha}^\dagger p_{k\beta}] \quad (35)$$

$$= \frac{2}{U}\tilde{g}_{rl} \sum_{k\beta} (\delta_{\sigma\alpha}\delta_{ri} - \delta_{\sigma\beta}\delta_{ik}) p_{r\alpha}^\dagger \tilde{g}_{kl} (\mathbf{S}_l \cdot \boldsymbol{\sigma}_{\alpha\beta}) p_{k\beta} \quad (36)$$

Inserting these results back we get two terms, the first is given by:

$$\mathbf{J}_{N\sigma}^1 = -i \sum_{ir} \mathbf{R}_i [n_i, h_r^1] \quad (37)$$

$$= -i \sum_{irl\alpha} \mathbf{R}_i \frac{t_{rl}}{2} \delta_{\alpha\sigma} \left( (\delta_{li} - \delta_{ri}) p_{l\sigma}^\dagger p_{r\sigma} + (\delta_{ri} - \delta_{li}) p_{r\sigma}^\dagger p_{l\sigma} \right) \quad (38)$$

$$= -i \left( \sum_{rl} \frac{t_{rl}}{2} (\mathbf{R}_l - \mathbf{R}_r) p_{l\sigma}^\dagger p_{r\sigma} + \sum_{rl} \frac{t_{rl}}{2} (\mathbf{R}_r - \mathbf{R}_l) p_{r\sigma}^\dagger p_{l\sigma} \right) \quad (39)$$

$$= -i \sum_{rl} t_{rl} (\mathbf{R}_l - \mathbf{R}_r) p_{l\sigma}^\dagger p_{r\sigma} \quad (40)$$

$$= -i \sum_{ij} t_{ij} (\mathbf{R}_i - \mathbf{R}_j) p_{i\sigma}^\dagger p_{j\sigma}. \quad (41)$$

For the second contribution, we get

$$\mathbf{J}_{N\sigma}^2 = -i \sum_{ir} \mathbf{R}_i [n_i, h_r^2] \quad (42)$$

$$= -i \sum_{irl\alpha} \mathbf{R}_i \frac{g_{rl}}{2} \left( \frac{2}{U} \tilde{g}_{rl} \sum_{k\beta} (\delta_{\sigma\alpha}\delta_{ri} - \delta_{\sigma\beta}\delta_{ik}) p_{r\alpha}^\dagger \tilde{g}_{kl} (\mathbf{S}_l \cdot \boldsymbol{\sigma}_{\alpha\beta}) p_{k\beta} + \frac{2}{U} \tilde{g}_{rl} \sum_{k\beta} (\delta_{\sigma\alpha}\delta_{li} - \delta_{\sigma\beta}\delta_{ik}) p_{l\alpha}^\dagger \tilde{g}_{kr} (\mathbf{S}_r \cdot \boldsymbol{\sigma}_{\alpha\beta}) p_{k\beta} \right) \quad (43)$$

$$= -i \sum_{rl\alpha} \frac{\tilde{g}_{rl}}{2} \left( \frac{4}{U} \sum_{k\beta} (\delta_{\sigma\alpha}\mathbf{R}_r - \delta_{\sigma\beta}\mathbf{R}_k) p_{r\alpha}^\dagger \tilde{g}_{kl} (\mathbf{S}_l \cdot \boldsymbol{\sigma}_{\alpha\beta}) p_{k\beta} \right) \quad (44)$$

$$= -i \sum_{rlk\beta\alpha} 2 \frac{\tilde{g}_{rl}\tilde{g}_{kl}}{U} (\delta_{\sigma\alpha}\mathbf{R}_r - \delta_{\sigma\beta}\mathbf{R}_k) p_{r\alpha}^\dagger (\mathbf{S}_l \cdot \boldsymbol{\sigma}_{\alpha\beta}) p_{k\beta} \quad (45)$$

Now we calculate the total current by summing over spins (index  $\sigma$ )

$$\mathbf{J}_N^2 = -i \sum_{rlk\beta\alpha} 2 \frac{\tilde{g}_{rl}\tilde{g}_{kl}}{U} (\mathbf{R}_r - \mathbf{R}_k) p_{r\alpha}^\dagger (\mathbf{S}_l \cdot \boldsymbol{\sigma}_{\alpha\beta}) p_{k\beta} \quad (46)$$

$$= -i \sum_{ilj\beta\alpha} 2 \frac{\tilde{g}_{il}\tilde{g}_{jl}}{U} (\mathbf{R}_i - \mathbf{R}_j) p_{i\alpha}^\dagger (\mathbf{S}_l \cdot \boldsymbol{\sigma}_{\alpha\beta}) p_{j\beta} \quad (47)$$

Then the total current is given by

$$\mathbf{J}_N = -i \sum_{ij\sigma} t_{ij} (\mathbf{R}_i - \mathbf{R}_j) p_{i\sigma}^\dagger p_{j\sigma} - i \sum_{ilj\beta\alpha} 2 \frac{g_{il} g_{jl}}{U} [1 - \lambda (X_{il} + X_{jl})] (\mathbf{R}_i - \mathbf{R}_j) p_{i\alpha}^\dagger (\mathbf{S}_l \cdot \boldsymbol{\sigma}_{\alpha\beta}) p_{j\beta} \quad (48)$$

In fourier space

$$\mathbf{J}_N^1(\omega_n) = \sum_{\mathbf{k}\sigma i\nu_n} v_{\mathbf{k}} p_{\mathbf{k}\sigma}^\dagger(i\nu_n) p_{\mathbf{k}\sigma}(\omega_n + \nu_n) \quad (49)$$

For the second part of the current operator

$$\mathbf{J}_N^2(\omega_n) = \mathbf{J}_N^{21}(\omega_n) + \mathbf{J}_N^{22}(\omega_n) \quad (50)$$

where we defined

$$\begin{aligned} \mathbf{J}_N^{21}(\omega_n) &= \sum_{\substack{\mathbf{k}\mathbf{k}'\beta\alpha ij \\ i\nu_l, i\nu_n}} \mathbf{f}_{ij}(\mathbf{k}, \mathbf{k}') p_{\mathbf{k}\alpha}^\dagger(i\nu_l) [\mathbf{S}_{\mathbf{k}-\mathbf{k}'}(i\nu_l - i\nu_n) \cdot \boldsymbol{\sigma}_{\alpha\beta}] p_{\mathbf{k}'\beta}(i\nu_n + i\omega_n) \\ \mathbf{J}_N^{22}(\omega_n) &= -\lambda \sum_{\substack{\mathbf{k}\mathbf{k}'\mathbf{q}\beta\alpha ij \\ i\nu_l, i\omega_m, i\nu_n}} \mathbf{f}_{ij}(\mathbf{k}, \mathbf{k}') [X_i(\mathbf{k} - \mathbf{k}' - \mathbf{q}, i\nu_l - i\omega_m - i\nu_n) \\ &\quad + X_j(\mathbf{k} - \mathbf{k}' - \mathbf{q}, i\nu_l - i\omega_m - i\nu_n)] p_{\mathbf{k}\alpha}^\dagger(i\nu_l) [\mathbf{S}_{\mathbf{q}}(i\omega_m) \cdot \boldsymbol{\sigma}_{\alpha\beta}] p_{\mathbf{k}'\beta}(i\nu_n + i\omega_n) \end{aligned}$$

and

$$\mathbf{f}_{ij}(\mathbf{k}, \mathbf{k}') = 2 \frac{g_{i0} g_{j0}}{U} (\mathbf{R}_i - \mathbf{R}_j) e^{-i\mathbf{k}\mathbf{R}_i + i\mathbf{k}'\mathbf{R}_j}. \quad (51)$$

#### IV. OUT OF PLANE CALCULATION

First we will do the calculation without vertex corrections. There is no contribution to the out of plane conductivity from the  $\mathbf{J}_N^1(\omega_n)$  component of the current since  $\mathbf{v}_{\mathbf{k}}$  is entirely in plane. To determine the out-of-plane conductivity we only need to take into account two contributions, namely

$$\sigma_{\perp}^1 = \frac{1}{\omega} \langle \mathbf{J}_N^{21}(\omega) \mathbf{J}_N^{21}(-\omega) \rangle$$

and

$$\sigma_{\perp}^2 = \frac{1}{\omega} \langle \mathbf{J}_N^{22}(\omega) \mathbf{J}_N^{22}(-\omega) \rangle$$

to leading order, since  $\langle X_i \rangle = 0$ . For the first of the two contributions above, we have

$$\begin{aligned} \Pi_1(i\omega_n) &= \frac{1}{\text{vol}} \int_0^\beta d\tau \sum_{\mathbf{k}\mathbf{q}} e^{i\omega_n \tau} \left| \sum_{ij} \mathbf{f}_{ij}(\mathbf{k}, \mathbf{k}') \right|^2 \langle T_\tau [\mathbf{S}_{\mathbf{k}-\mathbf{k}'}(\tau) \cdot \mathbf{S}_{\mathbf{k}'-\mathbf{k}}(0)] \rangle_\beta \mathcal{G}(\mathbf{k}, \tau) \mathcal{G}(\mathbf{k}', -\tau) \\ &= \frac{1}{\text{vol}} \int_0^\beta d\tau \sum_{\mathbf{k}\mathbf{k}'} \sum_{\alpha\gamma} e^{i\omega_n \tau} \Gamma^I(\mathbf{k}, \mathbf{k}') S_{\mathbf{k}-\mathbf{k}'}(\tau) \mathcal{G}(\mathbf{k}, \tau) \mathcal{G}(\mathbf{k}', -\tau) \end{aligned}$$

For the second, we have

$$\Pi_2(i\omega_n) = \frac{\lambda^2}{\text{vol}} \int_0^\beta d\tau \sum_{\mathbf{k}\mathbf{q}} \sum_{\alpha\gamma} e^{i\omega_n \tau} \Gamma^{II}(\mathbf{k}, \mathbf{k}') D(\mathbf{k} - \mathbf{k}' - \mathbf{q}, \tau) S_{\mathbf{q}}(\tau) \mathcal{G}(\mathbf{k}, \tau) \mathcal{G}(\mathbf{k}', -\tau)$$

with

$$\Gamma^{II}(\mathbf{k}, \mathbf{k}') = \sum_{ij i' j'} \mathbf{f}_{ij}(\mathbf{k}, \mathbf{k}') \cdot \overline{\mathbf{f}_{ij}}(\mathbf{k}, \mathbf{k}') (\delta_{ii'} + \delta_{ik'} + \delta_{ki'} + \delta_{kk'})$$

Now we will proceed with each correlation function individually

### A. First contribution: $\Pi_1(i\omega_n)$

We start by fourier transforming, which gives

$$\Pi_1(i\omega_n) = \sum_{\mathbf{k}\mathbf{k}'} \frac{1}{\beta^2} \sum_{i\mu_n, iz_r} \Gamma^I(\mathbf{k}, \mathbf{k}') S(\mathbf{k} - \mathbf{k}', i\mu_m - iz_r) \mathcal{G}(\mathbf{k}, i\mu_m) \mathcal{G}(\mathbf{k}', iz_r + i\omega_n)$$

where the two matsubara sums are fermionic. We focus now on the last term:

$$S(\mathbf{q}, i\nu_n) = - \int \frac{d\nu}{\pi} \frac{\chi''(\mathbf{q}, \nu)}{i\nu_n - \nu}, \quad (52)$$

$$\begin{aligned} & - \frac{1}{\beta} \sum_{iz_r} S(\mathbf{k} - \mathbf{k}', i\mu_m - iz_r) \mathcal{G}(\mathbf{k}', iz_r + i\omega_n) \\ & = - \int \frac{d\nu}{\pi} \chi''(\mathbf{k} - \mathbf{k}', \nu) \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \frac{1}{\beta} \sum_{iz_r} \frac{1}{\nu - (i\mu_m - iz_r)} \frac{1}{iz_r + i\omega_n - \epsilon_2} \end{aligned}$$

Evaluating the fermionic Matsubara sum

$$\frac{1}{\beta} \sum_{iz_r} \frac{1}{\nu - (i\mu_m - iz_r)} \frac{1}{iz_r + i\omega_n - \epsilon_2} = \int_{C_1} \frac{dz}{2\pi i} \frac{1}{z + \nu - i\mu_m} \frac{1}{z + i\omega_n - \epsilon_2} n_F(z) \quad (53)$$

$$= \int_{C_3} \frac{dz}{2\pi i} \frac{1}{z + \nu - i\mu_m} \frac{1}{z + i\omega_n - \epsilon_2} n_F(z) \quad (54)$$

$$= \frac{1}{\epsilon_2 - i\omega_n + \nu - i\mu_m} n_F(\epsilon_2 - i\omega_n) + \frac{1}{-(\nu - i\mu_m) + i\omega_n - \epsilon_2} n_F(-(\nu - i\mu_m)) \quad (55)$$

$$= -\frac{1}{i\omega_n + i\mu_n - \nu - \epsilon_2} n_F(\epsilon_2) + \frac{1}{i\omega_n + i\mu_n - \nu - \epsilon_2} n_F(i\mu_n - \nu) \quad (56)$$

$$= \frac{-n_B(-\nu) - n_F(\epsilon_2)}{i\omega_n + i\mu_n - \nu - \epsilon_2} \quad (57)$$

Inserting this back again into the current-current correlation function, we get:

$$\Pi_{zz}(i\omega_n) = - \sum_{\mathbf{k}\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\nu}{\pi} \int \frac{d\epsilon_2}{2\pi} \int \frac{d\epsilon_1}{2\pi} \chi''(\mathbf{k} - \mathbf{k}', \nu) A_{\text{Pd}}(\mathbf{k}, \epsilon_1) A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \frac{-n_B(-\nu) - n_F(\epsilon_2)}{i\omega_n + i\mu_n - \nu - \epsilon_2} \frac{1}{i\mu_r - \epsilon_1} \quad (58)$$

$$\begin{aligned} \Pi_{zz}(i\omega_n) &= \sum_{\mathbf{k}\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\nu}{\pi} \frac{d\epsilon_1}{2\pi} \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \chi''(\mathbf{k} - \mathbf{k}', \nu) (n_B(-\nu) + n_F(\epsilon_2)) \\ &\quad \times \frac{1}{\beta} \sum_{i\mu_n} \frac{1}{i\omega_n + i\mu_n - \nu - \epsilon_2} \frac{1}{i\mu_r - \epsilon_1} \end{aligned} \quad (59)$$

Evaluating the second fermionic Matsubara sum

$$\frac{1}{\beta} \sum_{i\nu_n} \frac{1}{i\omega_n + i\mu_n - \nu - \epsilon_2} \frac{1}{i\mu_r - \epsilon_1} = \int_{C_1} \frac{dz}{2\pi i} \frac{1}{i\omega_n + z - \nu - \epsilon_2} \frac{1}{z - \epsilon_1} n_F(z) \quad (60)$$

$$= - \int_{C_3} \frac{dz}{2\pi i} \frac{1}{i\omega_n + z - \nu - \epsilon_2} \frac{1}{z - \epsilon_1} n_F(z) \quad (61)$$

$$= \left[ \frac{1}{i\omega_n + \epsilon_1 - \nu - \epsilon_2} n_F(\epsilon_1) + \frac{1}{-(i\omega_n - \nu - \epsilon_2) - \epsilon_1} n_F(-(i\omega_n - \nu - \epsilon_2)) \right] \quad (62)$$

$$= - \frac{n_F(\epsilon_2 + \nu) - n_F(\epsilon_1)}{i\omega_n - \nu - \epsilon_2 + \epsilon_1} \quad (63)$$

Putting this back into  $\Pi$ , we get

$$\Pi_{zz}(i\omega_n) = - \sum_{\mathbf{k}\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\nu}{\pi} \frac{d\epsilon_1}{2\pi} \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \chi''(\mathbf{k} - \mathbf{k}', \nu) (n_B(-\nu) + n_F(\epsilon_2)) \frac{n_F(\epsilon_2 + \nu) - n_F(\epsilon_1)}{i\omega_n - \nu - \epsilon_2 + \epsilon_1} \quad (64)$$

Analytically continuing to real frequencies and taking the imaginary part and using the Sokhotski-Plemelj formula:

$$\frac{1}{\omega + \zeta \pm i\delta} = P \frac{1}{\omega + \zeta} \mp i\pi \delta(\omega + \zeta), \quad (65)$$

we get

$$\begin{aligned} \text{Im}\Pi_{zz}(\omega) &= \pi \sum_{\mathbf{k}\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\nu}{\pi} \frac{d\epsilon_1}{2\pi} \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \chi''(\mathbf{k} - \mathbf{k}', \nu) \\ &\quad \times (n_B(-\nu) + n_F(\epsilon_2)) [n_F(\epsilon_2 + \nu) - n_F(\epsilon_1)] \delta(\omega - \nu - \epsilon_2 + \epsilon_1) \\ &= \pi \sum_{\mathbf{k}\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_1}{2\pi} \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \chi''(\mathbf{k} - \mathbf{k}', \omega - \epsilon_2 + \epsilon_1) \\ &\quad \times (n_B(\epsilon_2 - \omega - \epsilon_1) + n_F(\epsilon_2)) [n_F(\epsilon_2 + (\omega - \epsilon_2 + \epsilon_1)) - n_F(\epsilon_1)] \\ &= \pi \sum_{\mathbf{k}\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_1}{2\pi} \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \chi''(\mathbf{k} - \mathbf{k}', \omega - \epsilon_2 + \epsilon_1) \\ &\quad \times (n_B(\epsilon_2 - \omega - \epsilon_1) + n_F(\epsilon_2)) [n_F(\omega + \epsilon_1) - n_F(\epsilon_1)] \end{aligned}$$

Where in the second step we performed the integral over  $\nu$ . To calculate the optical conductivity we simply divide by the frequency

$$\begin{aligned} \text{Re}\sigma_{zz}(\omega) &= \frac{\text{Im}\Pi_{zz}(\omega)}{\omega} \\ &= \sum_{\mathbf{k}\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_1}{2\pi} \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \chi''(\mathbf{k} - \mathbf{k}', \omega - \epsilon_2 + \epsilon_1) (n_B(\epsilon_2 - \omega - \epsilon_1) + n_F(\epsilon_2)) \\ &\quad \times \frac{[n_F(\omega + \epsilon_1) - n_F(\epsilon_1)]}{\omega} \end{aligned}$$

Taking the D.C. limit, we get

$$\sigma_{\perp}^1 = e^2 \sum_{\mathbf{k}\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_1}{2\pi} \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \chi''(\mathbf{k} - \mathbf{k}', \epsilon_1 - \epsilon_2) (n_B(\epsilon_1 - \epsilon_2) + n_F(-\epsilon_2)) \left( -\frac{\partial n_f}{\partial \epsilon_1} \right) \quad (66)$$

$$= e^2 \sum_{\mathbf{k}} \int \frac{d\epsilon_1}{2\pi} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) A_{\perp,1}(\mathbf{k}, \epsilon_1) \left( -\frac{\partial n_f}{\partial \epsilon_1} \right) \quad (67)$$

where we defined

$$A_{\perp,1}(\mathbf{k}, \epsilon_1) = \sum_{\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \chi''(\mathbf{k} - \mathbf{k}', \epsilon_1 - \epsilon_2) (n_B(\epsilon_1 - \epsilon_2) + n_F(-\epsilon_2))$$

where we reinstated the electron charge in the last step of the calculation.

### B. second contribution: $\Pi_2(i\omega_n)$

Fourier transforming we have

$$\Pi_1(i\omega_n) = \lambda^2 \sum_{\mathbf{k}\mathbf{k}'\mathbf{q}} \frac{1}{\beta^3} \sum_{i\mu_n, iz_n, i\nu_n} \Gamma^{II}(\mathbf{k}, \mathbf{k}') D(\mathbf{k} - \mathbf{k}' - \mathbf{q}, i\mu_n - iz_n - i\nu_n) S(\mathbf{q}, i\nu_n) \mathcal{G}(\mathbf{k}, i\mu_n) \mathcal{G}(\mathbf{k}', iz_n + i\omega_n)$$

where we have to perform two fermionic matsubara sums and a bosonic matubara sum. We start by performing the bosonic sums

$$\begin{aligned} \frac{1}{\beta} \sum_{i\nu_n} D(\mathbf{k} - \mathbf{k}' - \mathbf{q}, i\mu_n - iz_n - i\nu_n) S(\mathbf{q}, i\nu_n) &= \frac{1}{\beta} \int \frac{d\nu}{\pi} \chi''(\mathbf{q}, \nu) \sum_{i\nu_n} \frac{1}{(i\mu_n - iz_n - i\nu_n)^2 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}^2} \frac{1}{i\nu_n - \nu} \\ &= \frac{1}{\beta} \int \frac{d\nu}{\pi} \chi''(\mathbf{q}, \nu) \frac{\hbar}{2M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} \\ &\quad \left[ \sum_{i\nu_n} \frac{1}{i\mu_n - iz_n - i\nu_n - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} \frac{1}{i\nu_n - \nu} \right. \\ &\quad \left. - \sum_{i\nu_n} \frac{1}{i\mu_n - iz_n - i\nu_n + \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} \frac{1}{i\nu_n - \nu} \right] \end{aligned}$$

these yield the result

$$\begin{aligned} \frac{1}{\beta} \sum_{i\nu_n} D(\mathbf{k} - \mathbf{k}' - \mathbf{q}, i\mu_n - iz_n - i\nu_n) S(\mathbf{q}, i\nu_n) &= \frac{\hbar}{2M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} \frac{1 + n_b(\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) + n_b(\nu)}{i\mu_n - iz_n - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}} - \nu} \\ &\quad - \frac{\hbar}{2M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} \frac{1 + n_b(-\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) + n_b(\nu)}{i\mu_n - iz_n + \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}} - \nu} \end{aligned}$$

then, we can use the same results as above and we get the following

$$\sigma_{\perp}^2 = e^2 \sum_{\mathbf{k}} \int \frac{d\epsilon_1}{2\pi} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) A_{\perp,2}(\mathbf{k}, \epsilon_1) \left( -\frac{\partial n_f}{\partial \epsilon_1} \right) \quad (68)$$

with

$$A_{\perp,2}(\mathbf{k}, \epsilon_1) = \sum_{\mathbf{q}\mathbf{k}'} \frac{\hbar\lambda^2}{2M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} \Gamma^{II}(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}', \epsilon_2) [n_B(\epsilon_1 - \epsilon_2) + n_F(-\epsilon_2)] \quad (69)$$

$$\times \left\{ [1 + n_b(\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) + n_b(\epsilon_1 - \epsilon_2 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}})] \chi''(\mathbf{q}, \epsilon_1 - \epsilon_2 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) \right. \quad (70)$$

$$\left. + [1 + n_b(\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) + n_b(\epsilon_2 - \epsilon_1 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}})] \chi''(\mathbf{q}, \epsilon_2 - \epsilon_1 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) \right\} \quad (71)$$

here the first two lines in the rate  $\gamma$  corresponds to emission of a phonon and the second two lines corresponds to absorption.

## V. SELF ENERGY

The self energy in the RPA approximation has the following self consistent equation

$$\begin{aligned}\Sigma(\mathbf{k}, i\omega_n) &= \sum_{\mathbf{k}'} \int_0^\beta d\tau e^{i\omega_n \tau} \left[ \langle \langle \overline{U_{\mathbf{k}\mathbf{k}'}^{\alpha\beta}(\tau)} U_{\mathbf{k}\mathbf{k}'}^{\alpha\beta}(0) \rangle \rangle + \langle \langle \overline{W_{\mathbf{k}\mathbf{k}'}^{\alpha\beta}(\tau)} W_{\mathbf{k}\mathbf{k}'}^{\alpha\beta}(0) \rangle \rangle \right] \mathcal{G}(\mathbf{k}', \tau) \\ &= \frac{1}{\beta} \sum_{\mathbf{k}'} \sum_{i\nu_n} \left[ \Lambda_I(\mathbf{k}, \mathbf{k}') S(\mathbf{k} - \mathbf{k}', i\omega_n - i\nu_n) + \lambda^2 \frac{1}{\beta} \sum_{\mathbf{q}} \sum_{i\mu_n} \Lambda_{II}(\mathbf{k}, \mathbf{k}') \frac{1}{M} \frac{1}{(i\omega_n - i\nu_n - i\mu_n)^2 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}^2} S(\mathbf{q}, i\mu_n) \right] \\ &\quad \times \mathcal{G}(\mathbf{k}', i\nu_n)\end{aligned}$$

where the form factors are again:

$$\Lambda_I(\mathbf{k}, \mathbf{k}') = \frac{1}{N} \sum_{ik i' k'} g_{ik}(\mathbf{k}, \mathbf{k}') \overline{g_{i'k'}}(\mathbf{k}, \mathbf{k}') \quad (72)$$

$$\Lambda_{II}(\mathbf{k}, \mathbf{k}') = \frac{1}{N} \sum_{ik i' k'} g_{ik}(\mathbf{k}, \mathbf{k}') \overline{g_{i'k'}}(\mathbf{k}, \mathbf{k}') (\delta_{ii'} + \delta_{ik'} + \delta_{ki'} + \delta_{kk'}) \quad (73)$$

Simplifying the self energy in the limit where  $T \gg \omega_D$ , we have

$$\begin{aligned}\Sigma(\mathbf{k}, i\omega_n) &= \frac{\lambda^2}{\beta^2} \sum_{\mathbf{k}'\mathbf{q}} \sum_{i\nu_n} \left[ \frac{\Lambda_{II}(\mathbf{k}, \mathbf{k}')}{M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}^2} S(\mathbf{q}, i\omega_n - i\nu_n) \right] \mathcal{G}(\mathbf{k}', i\nu_n) \\ &= \frac{\lambda^2}{\beta^2} \sum_{\mathbf{k}'\mathbf{q}} \sum_{i\nu_n} \left[ \frac{\Lambda_{II}(\mathbf{k}, \mathbf{k}')}{M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}^2} \int \frac{d\nu}{\pi} \frac{-\chi''(\mathbf{q}, \nu)}{i\omega_n - i\nu_n - \nu} \right] \mathcal{G}(\mathbf{k}', i\nu_n) \\ &= \frac{\lambda^2}{\beta} \int \frac{d\nu}{\pi} \int \frac{d\epsilon}{2\pi} \sum_{\mathbf{k}'\mathbf{q}} \left[ \frac{\Lambda_{II}(\mathbf{k}, \mathbf{k}')}{M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}^2} \chi''(\mathbf{q}, \nu) A_{\text{Pd}}(\mathbf{k}', \epsilon) \right] \frac{1}{\beta} \sum_{i\nu_n} \frac{-1}{i\omega_n - i\nu_n - \nu} \frac{1}{i\nu_n - \epsilon} \\ &= \frac{\lambda^2}{\beta} \int \frac{d\nu d\epsilon}{2\pi^2} \sum_{\mathbf{k}'\mathbf{q}} \frac{\Lambda_{II}(\mathbf{k}, \mathbf{k}')}{M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}^2} \chi''(\mathbf{q}, \nu) A_{\text{Pd}}(\mathbf{k}', \epsilon) \left[ -\frac{n_f(\epsilon) + n_b(-\nu)}{i\omega_n - \nu - \epsilon} \right]\end{aligned}$$

in the hight temperature limit, we then have

$$\Sigma(\mathbf{k}, i\omega_n) = \frac{\lambda^2}{\beta} \int \frac{d\nu d\epsilon}{2\pi^2} \sum_{\mathbf{k}'\mathbf{q}} \frac{\Lambda_{II}(\mathbf{k}, \mathbf{k}')}{M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}^2} \chi''(\mathbf{q}, \nu) A_{\text{Pd}}(\mathbf{k}', \epsilon) \frac{n_f(-\epsilon) + n_b(\nu)}{i\omega_n - \nu - \epsilon} \quad (74)$$

where we worked at leading order dismissing any back reaction on the phonon and spin subsystems.

## VI. SIMPLIFIED COMPARISSON: NO VERTEX CORRECTIONS.

Performing the calculation without any vertex corrections yields the following conductivities:

$$\sigma_{\perp} = e^2 \sum_{\mathbf{k}} \int \frac{d\epsilon_1}{2\pi} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) A_{\perp}(\mathbf{k}, \epsilon_1) \left( -\frac{\partial n_f}{\partial \epsilon_1} \right) \quad (75)$$

$$\sigma_{\parallel} = e^2 \int \frac{d\epsilon_1}{2\pi} \sum_{\mathbf{k}} v_{\mathbf{k}}^2 A_{\text{Pd}}^2(\mathbf{k}, \epsilon_1) \left( -\frac{\partial n_f}{\partial \epsilon_1} \right). \quad (76)$$

The difference lies in the inclusion of the modified spectral function  $A_{\perp}(\mathbf{k}, \epsilon_1)$  for the out of plane conductivity which is related to the form of the full current operator. In particular, we have

$$A_{\perp}(\mathbf{k}, \epsilon_1) = A_{\perp,1}(\mathbf{k}, \epsilon_1) + A_{\perp,2}(\mathbf{k}, \epsilon_1) \quad (77)$$

with the modified spectral functions

$$A_{\perp,1}(\mathbf{k}, \epsilon_1) = \sum_{\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \chi''(\mathbf{k} - \mathbf{k}', \epsilon_1 - \epsilon_2) (n_B(\epsilon_1 - \epsilon_2) + n_F(-\epsilon_2))$$

and

$$A_{\perp,2}(\mathbf{k}, \epsilon_1) = \sum_{\mathbf{q}\mathbf{k}'} \frac{\hbar\lambda^2}{2M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} \Gamma^{II}(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}', \epsilon_2) [n_B(\epsilon_1 - \epsilon_2) + n_F(-\epsilon_2)] \quad (78)$$

$$\times \left\{ \left[ 1 + n_b(\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) + n_b(\epsilon_1 - \epsilon_2 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) \right] \chi''(\mathbf{q}, \epsilon_1 - \epsilon_2 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) \right. \quad (79)$$

$$\left. + \left[ 1 + n_b(\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) + n_b(\epsilon_2 - \epsilon_1 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) \right] \chi''(\mathbf{q}, \epsilon_2 - \epsilon_1 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) \right\} \quad (80)$$

Now we consider the limit in which  $T \ll \epsilon_F$  in this limit we have

$$\sigma_{\perp} = e^2 \sum_{\mathbf{k}} A_{\text{Pd}}(\mathbf{k}, 0) A_{\perp}(\mathbf{k}, 0) \quad (81)$$

$$\sigma_{\parallel} = e^2 \sum_{\mathbf{k}} v_{\mathbf{k}}^2 A_{\text{Pd}}^2(\mathbf{k}, 0). \quad (82)$$

With the spectral function is given by

$$A_{\text{Pd}}(\mathbf{k}, \epsilon_1) = \frac{2\Sigma''(\mathbf{k}, \epsilon_1)}{(\epsilon_1 - \xi_{\mathbf{k}} - \Sigma'(\mathbf{k}, \epsilon_1))^2 + \Sigma''(\mathbf{k}, \epsilon_1)^2}$$

and in the high temperature limit, we have:

$$A_{\perp,2}(\mathbf{k}, \epsilon_1) = k_b T \sum_{\mathbf{q}\mathbf{k}'} \frac{\lambda^2}{2M\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} \Gamma^{II}(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}', \epsilon_2) [n_B(\epsilon_2 - \epsilon_1) + n_F(\epsilon_2)] \quad (83)$$

$$\times \left\{ \left[ \frac{1}{\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} + \frac{1}{\epsilon_2 - \epsilon_1 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} \right] \chi''(\mathbf{q}, \epsilon_2 - \epsilon_1 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) \right. \quad (84)$$

$$\left. + \left[ \frac{1}{\omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} + \frac{1}{\epsilon_1 - \epsilon_2 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}} \right] \chi''(\mathbf{q}, \epsilon_1 - \epsilon_2 - \omega_{\mathbf{k}-\mathbf{k}'-\mathbf{q}}) \right\} \quad (85)$$

clearly more care needs to go into the calculation accounting for the self energy and the vertex corrections consistently to obey the Ward Identity. Doing this should bring about the correct temperature dependence for  $\sigma_{\perp}$ .

## VII. PHONON ONLY

The derivation for  $\Pi_1$  would be exactly the same if the coupling was mediated by phonons, the only things that change then are the form factor and the spectral function, all of the other results are unaltered. Starting from the result above, we have

$$\sigma_{\perp}^1 = e^2 \sum_{\mathbf{k}} \int \frac{d\epsilon_1}{2\pi} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) A_{\perp,1}(\mathbf{k}, \epsilon_1) \left( -\frac{\partial n_f}{\partial \epsilon_1} \right) \quad (86)$$

where we defined

$$A_{\perp,1}(\mathbf{k}, \epsilon_1) = \sum_{\mathbf{k}'} M(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \chi''(\mathbf{k} - \mathbf{k}', \epsilon_1 - \epsilon_2) (n_B(\epsilon_1 - \epsilon_2) + n_F(-\epsilon_2))$$

where  $M(\mathbf{k}, \mathbf{k}')$  is a different form factor. Taking the spectral function to be

$$\chi''(\mathbf{k} - \mathbf{k}', \epsilon_1 - \epsilon_2) = 2\pi \frac{1}{M\omega_0} (\delta(\epsilon_1 - \epsilon_2 - \omega_0) - \delta(\epsilon_1 - \epsilon_2 + \omega_0))$$

Then we have

$$\begin{aligned} A_{\perp,1}(\mathbf{k}, \epsilon_1) &= \frac{1}{M\omega_0} 2\pi \sum_{\mathbf{k}'} M(\mathbf{k}, \mathbf{k}') A_{\text{Pd}}(\mathbf{k}', \epsilon_1 - \omega_0) (n_B(\omega_0) + n_F(\omega_0 - \epsilon_1)) \\ &\quad - \frac{1}{M\omega_0} 2\pi \sum_{\mathbf{k}'} M(\mathbf{k}, \mathbf{k}') A_{\text{Pd}}(\mathbf{k}', \epsilon_1 + \omega_0) (n_B(-\omega_0) + n_F(-\omega_0 - \epsilon_1)) \\ &\sim \frac{1}{M\omega_0} 2\pi \sum_{\mathbf{k}'} M(\mathbf{k}, \mathbf{k}') A_{\text{Pd}}(\mathbf{k}', \epsilon_1) (n_B(\omega_0) + n_F(\omega_0 - \epsilon_1) - n_B(-\omega_0) - n_F(-\omega_0 - \epsilon_1)) \\ &\sim \frac{1}{M\omega_0} 2\pi \sum_{\mathbf{k}'} M(\mathbf{k}, \mathbf{k}') A_{\text{Pd}}(\mathbf{k}', \epsilon_1) (n_B(\omega_0) - n_B(-\omega_0)) \\ &= \frac{1}{M\omega_0} 2\pi \sum_{\mathbf{k}'} M(\mathbf{k}, \mathbf{k}') A_{\text{Pd}}(\mathbf{k}', \epsilon_1) (2n_B(\omega_0) + 1) \end{aligned}$$

then to leading order in  $\omega_0/T$  we have

$$A_{\perp,1}(\mathbf{k}, \epsilon_1) \sim 4\pi \frac{T}{M\omega_0^2} \sum_{\mathbf{k}'} M(\mathbf{k}, \mathbf{k}') A_{\text{Pd}}(\mathbf{k}', \epsilon_1)$$

So that at high temperatures, we have

$$\begin{aligned} \sigma_{\perp}^1 &= \frac{e^2 4\pi T}{M\omega_0^2} \sum_{\mathbf{k}\mathbf{k}'} M(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_1}{2\pi} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) A_{\text{Pd}}(\mathbf{k}', \epsilon_1) \left( -\frac{\partial n_f}{\partial \epsilon_1} \right) \\ &\sim 2 \frac{e^2 T}{M\omega_0^2} \sum_{\mathbf{k}\mathbf{k}'} M(\mathbf{k}, \mathbf{k}') A_{\text{Pd}}(\mathbf{k}, 0) A_{\text{Pd}}(\mathbf{k}', 0) \end{aligned}$$

Then if we take the high temperature limit with

$$\begin{aligned} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) &= \frac{2\Sigma''(\mathbf{k}, \epsilon_1)}{(\epsilon_1 - \xi_{\mathbf{k}} - \Sigma'(\mathbf{k}, \epsilon_1))^2 + \Sigma''(\mathbf{k}, \epsilon_1)^2} \\ &\sim \frac{1}{\Sigma''(\mathbf{k}, \epsilon_1)} \sim \frac{1}{T} \end{aligned}$$

we would have

$$\sigma_{\perp}^1 \sim \frac{1}{T}.$$

### VIII. MAGNETOELASTIC HIGH T APPROXIMATION

First for the magnetic only, we have

$$\begin{aligned}
A_{\perp,1}(\mathbf{k}, \epsilon_1) &= \sum_{\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \chi''(\mathbf{k} - \mathbf{k}', \epsilon_1 - \epsilon_2) (n_B(\epsilon_1 - \epsilon_2) + n_F(-\epsilon_2)) \\
&= \sum_{\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \beta(\epsilon_1 - \epsilon_2) \theta(J - |\epsilon_1 - \epsilon_2|) (n_B(\epsilon_1 - \epsilon_2) + n_F(-\epsilon_2)) \\
&= \sum_{\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \beta(\epsilon_1 - \epsilon_2) \theta(J - |\epsilon_1 - \epsilon_2|) \left( \frac{1}{\epsilon_1 - \epsilon_2} + \frac{1}{-\epsilon_2} \right) \\
&= \sum_{\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \theta(J - |\epsilon_1 - \epsilon_2|) \left( \frac{-\epsilon_2 + \epsilon_1 - \epsilon_2}{-\epsilon_2} \right) \\
&= \sum_{\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \theta(J - |\epsilon_1 - \epsilon_2|) \left( \frac{2\epsilon_2 - \epsilon_1}{\epsilon_2} \right)
\end{aligned}$$

then we have

$$\sigma_{\perp} = e^2 \sum_{\mathbf{k}} \int \frac{d\epsilon_1}{2\pi} A_{\text{Pd}}(\mathbf{k}, \epsilon_1) \sum_{\mathbf{k}'} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \theta(J - |\epsilon_1 - \epsilon_2|) \left( \frac{2\epsilon_2 - \epsilon_1}{\epsilon_2} \right) \left( -\frac{\partial n_f}{\partial \epsilon_1} \right) \quad (87)$$

$$e^2 \sum_{\mathbf{k}' \mathbf{k}} \Gamma^I(\mathbf{k}, \mathbf{k}') \int \frac{d\epsilon_2}{2\pi} A_{\text{Pd}}(\mathbf{k}, 0) A_{\text{Pd}}(\mathbf{k}', \epsilon_2) \theta(J - |\epsilon_2|) \quad (88)$$